



Getting to know your home network...

(... and Network tools for the rest of
the module)

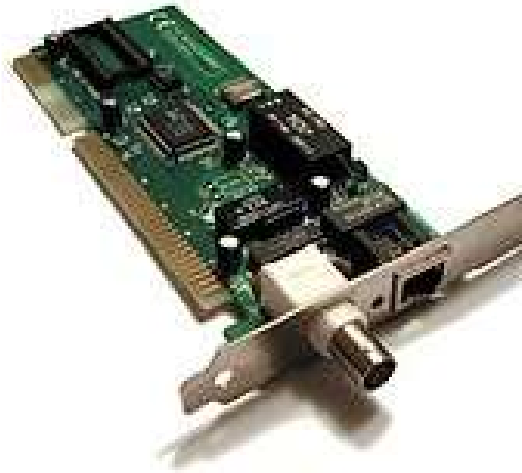
Agenda

- Adapters/NICs/Connections
- Virtual Machine Adapters
 - Bridging and other stuff
- Home Network Analysis
 - Nmap
 - Wireshark
 - Packet Tracer



What's a Network Adapter?

- AKA **Network Interface Controller(NIC)**, LAN adapter, **Physical Network Interface**.
- Usually the piece of hardware built in to a device.
- Interconnection point between a device and a network.
- Actually does not need to be a physical thing
 - E.g. Your Virtual Machine has a Virtual Adapter
- This provides a base for a full network protocol stack (More later)
- On some Operating Systems, known as Network Connection/Device.



Virtual Box

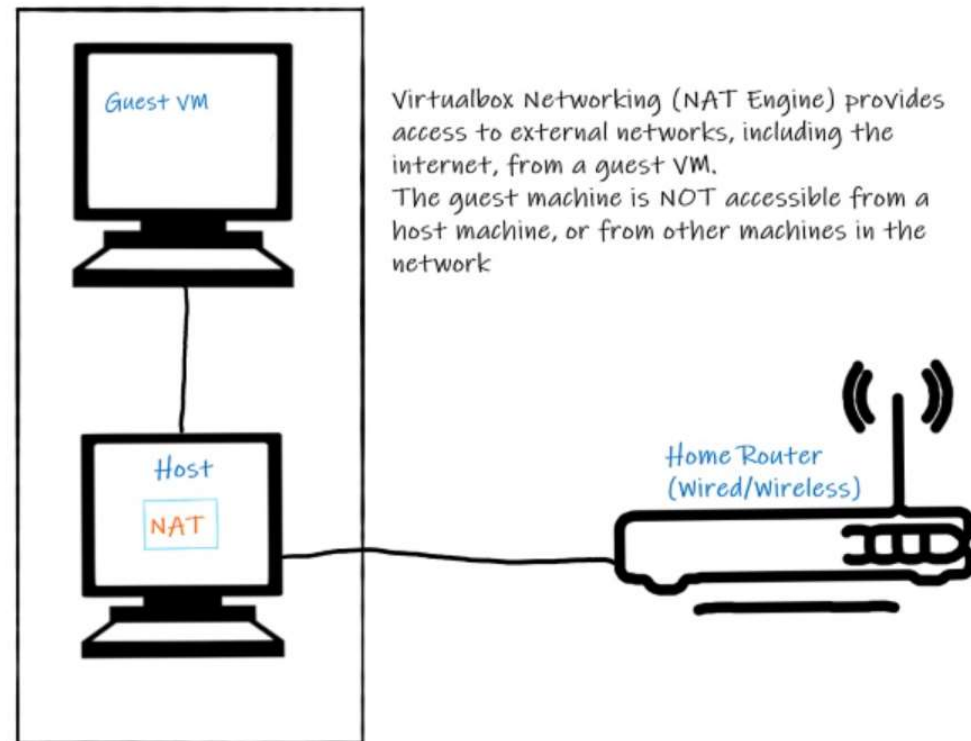
Bridged Adapter

Shared Folders

Other Stuff...

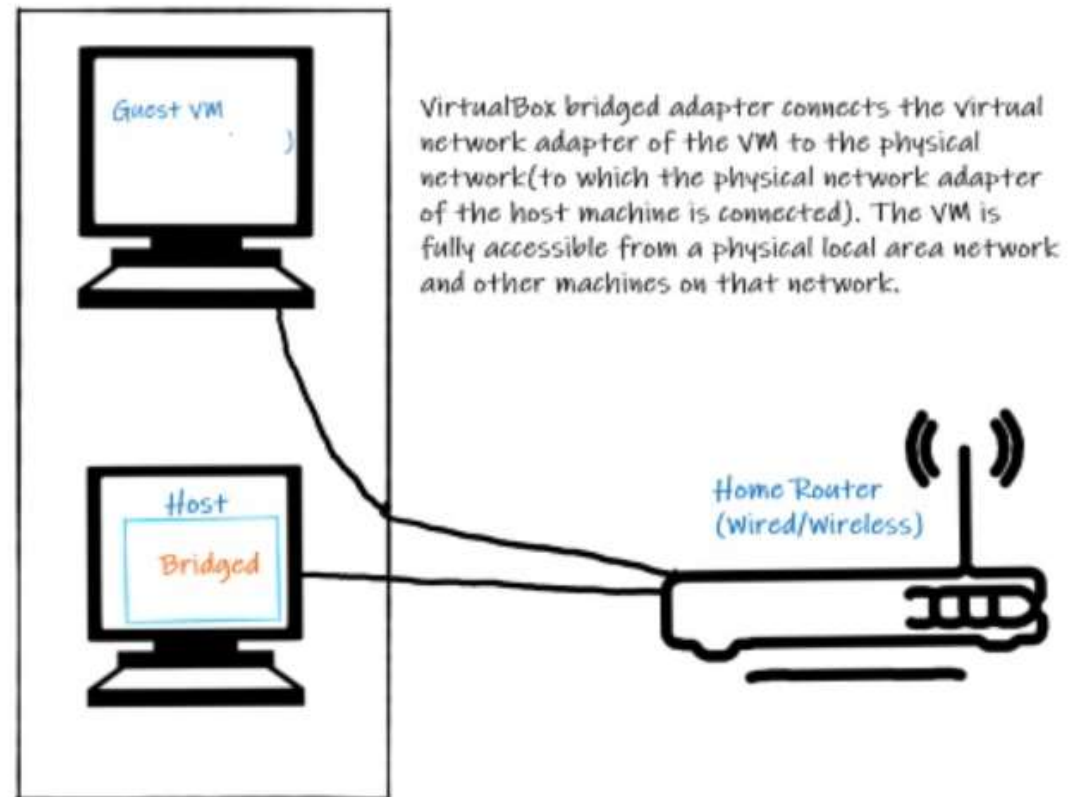
The Current VM setup

- Up to now (using something called NAT)
- Problem: Cannot “see” home network from Guest VM



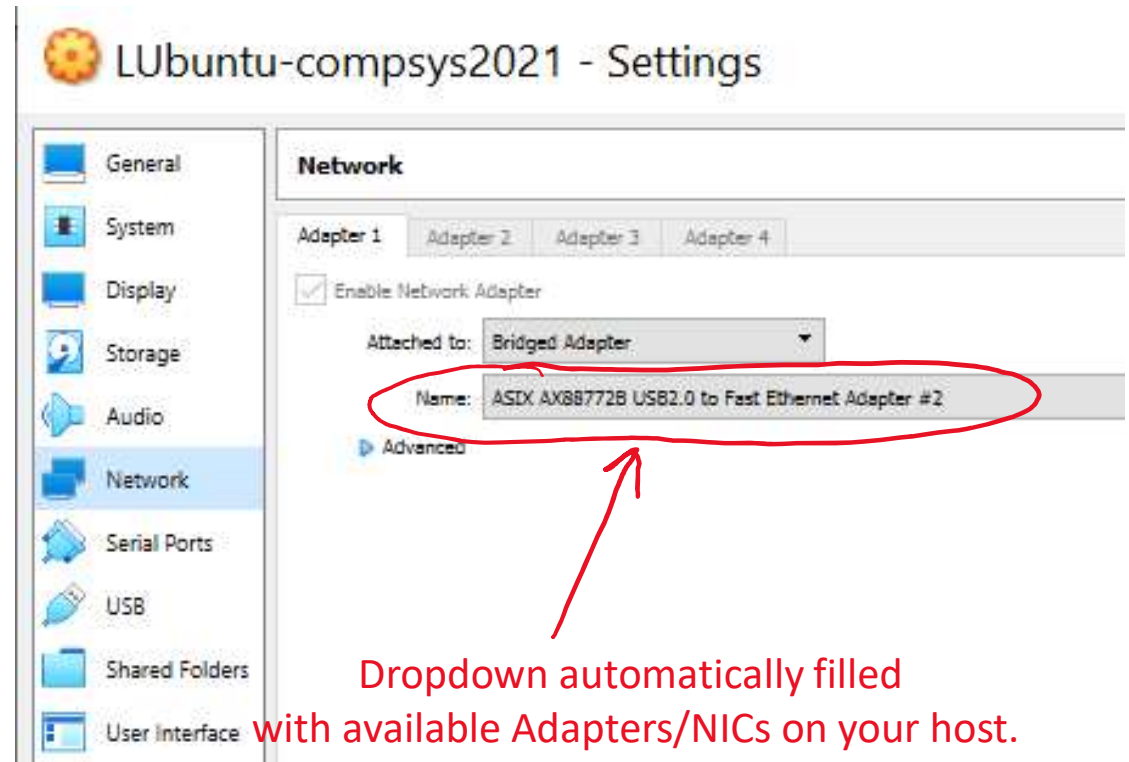
Preferred VM Setup

- Bridged Adapter
 - Virtual Box connects the VM directly to the physical network via hosts network adapters
- How does that work?
 - <https://www.virtualbox.org/manual/ch06.html>



VirtualBox Bridged Adapter

- To configure it, you need to select the Physical Adapter on the host the you want to use.
- **REMEMBER: if you change from wired to wireless connection (or vice versa) you'll have to change this**



Dropdown automatically filled
with available Adapters/NICs on your host.
Select the one you're using
to connect to your network

What if Bridged Adapter does not work...

- Some systems may not support bridging(rare). (NIC hardware/Drivers)
- You can set up your host with the tools required to do the labs (preferable to use VM though as it matches labs.)
 - You'll need Packet Tracer, Wireshark, Python 3.x, Node



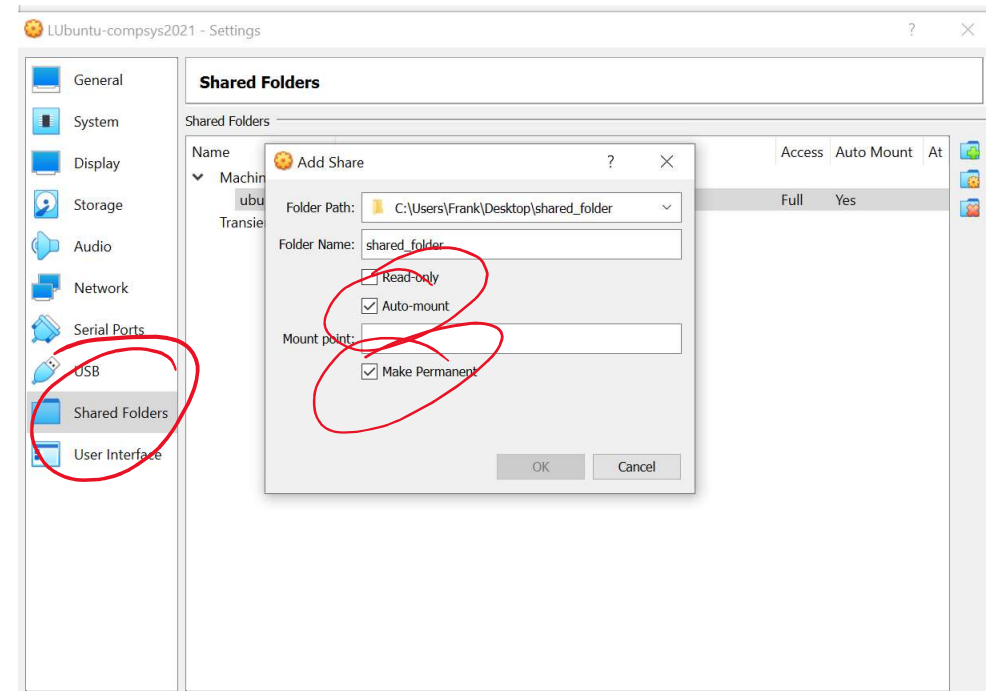
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Shared Folder between VirtualBox guest and host.

- Handy for moving files from guest to host/host to guest
- In Virtual Box Manager, add the folder you want to share
- Run the following at the command line in the VM:

```
sudo adduser compsys vboxsf
```

- Reboot the VM!



<https://www.howtogeek.com/187703/how-to-access-folders-on-your-host-machine-from-an-ubuntu-virtual-machine-in-virtualbox/>

Nmap

Introduction

What's Nmap



Short for Network Mapper



Utility for scanning a network



Used for IT auditing,
Device/Asset discovery,
Security profiling, Hacking!

Nmap Usage – What's out there.

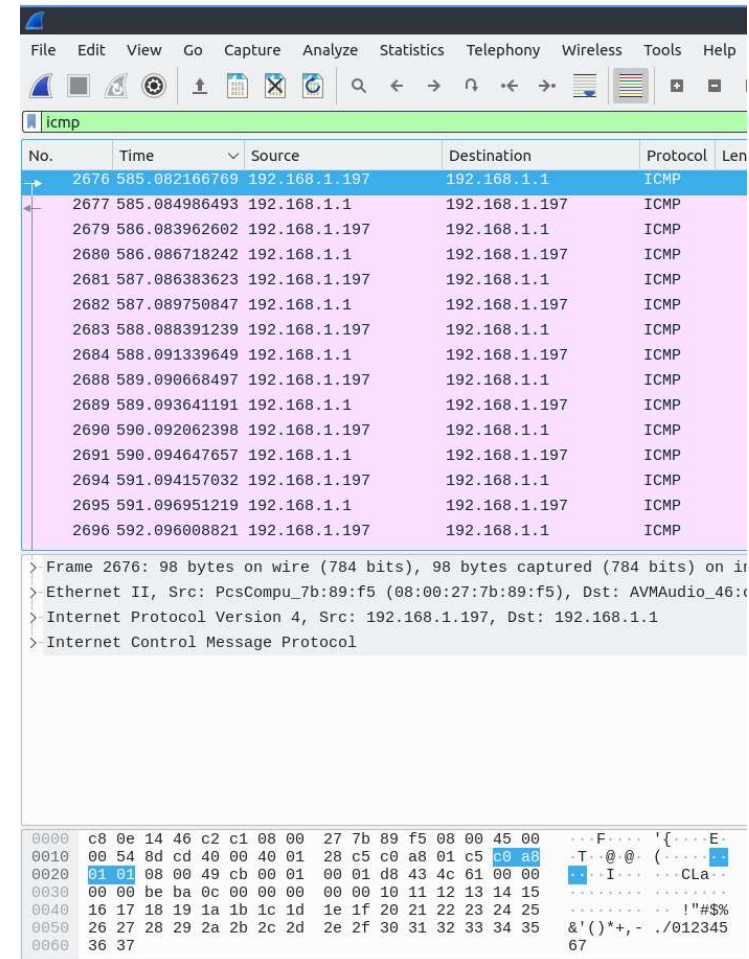
- Find out what's on your network:
`sudo nmap -sn 192.168.1.0/24`
- Detect Operating Systems:
`sudo nmap -O 192.168.1.234`
- Check open ports (more about this later)
`sudo nmap -sT 192.168.1.0/24`
- Check it all:
 - `sudo nmap -sA 192.168.1.123`
- Command line tool:
 - You can use all the scripting goodness from previous section to do interesting stuff!
- DEMO.

```
root@kali:~# nmap -sn 192.168.56.0/24

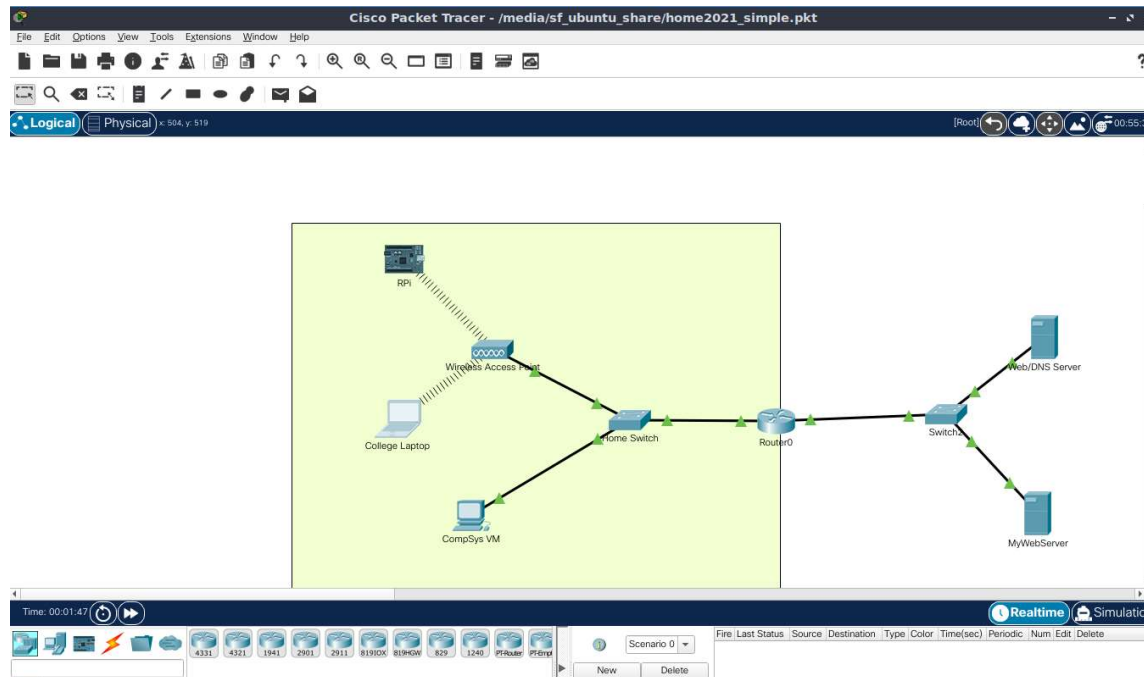
Starting Nmap 7.30 ( https://nmap.org ) at 2016-11-02 20:28 EDT
Nmap scan report for 192.168.56.1
Host is up (0.00041s latency).
MAC Address: 0A:00:27:00:00:00 (Unknown)
Nmap scan report for 192.168.56.100
Host is up (0.00018s latency).
MAC Address: 08:00:27:98:62:C4 (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.102
Host is up (0.00032s latency).
MAC Address: 08:00:27:34:58:53 (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.101
Host is up.
Nmap done: 256 IP addresses (4 hosts up)
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```

Wireshark

- network protocol analyser
 - see what's happening on your network at a microscopic level
- <https://www.wireshark.org/>



Packet Tracer



- Network Simulator
- Visual tool to create and run network topologies
- Includes IoT Devices.